

Simon Minami

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EDUCATION

Carnegie Mellon University

MS in Computer Vision

Pittsburgh, PA

Graduation Date: December 2026

- Courses: Advanced Computer Vision, Visual Learning and Recognition, Learning for 3D Vision

Tufts University

BS in Data Science (GPA: 3.7)

Medford, MA

Graduation Date: August 2024

- Awards: Magna Cum Laude, Dean's List | Minors: Economics and Mathematics

WORK EXPERIENCE

CMU Robotics Institute

Graduate Student Researcher

Pittsburgh, PA

January 2026 - Present

- Designed an end-to-end multiview (exocentric and egocentric) capture rig and preprocessing pipeline, automating camera calibration, temporal synchronization, segmentation, and image undistortion for large-scale 4D reconstruction datasets.
- Developed a joint 4D hand-object reconstruction pipeline integrating dynamic Gaussian Splatting, hand pose estimation, and contact reasoning to reconstruct deformable clay interactions under severe hand-object occlusion.

GE Aerospace

AI / Computer Vision Intern

Niskayuna, NY

May - August 2026

- Engineered a cross-source SLAM-to-CAD point cloud registration pipeline for autonomous engine inspection robotics, reducing localization error by 30%.
- Boosted point cloud correspondence inlier rate by 20% in sparse, low-overlap, and noisy sensor environments by investigating hybrid deep learning-based and classical matching methods.

CBB Analytics

Computer Vision Engineer

Remote

June 2024 - August 2025

- Increased spatial accuracy of player movements by 15% by developing a pipeline to extract precise x and y coordinates, using advanced machine learning models in PyTorch.
- Improved player tracking metrics by 10% in basketball game footage by developing and implementing a basketball player tracking model using advanced multi-object tracking techniques.
- Developed a homography estimation model mapping broadcast video to a standardized court view, enabling precise spatial analytics and event tracking.
- Fine-tuned a YOLO model to detect scoreboard components and applied OCR with custom preprocessing (binarization, thinning) for text extraction, enabling detailed, frame-level game event analysis and data-driven sports insights.

SKILLS & INTERESTS

Skills: Python (PyTorch, OpenCV, Open3D, NumPy, Pandas, Scikit-learn), C++, SQL, Gaussian Splatting, 3D Reconstruction, Multi-View Geometry, Camera Calibration, Blender, Google Cloud Platform (GCS, BigQuery), Git, MATLAB

Interests: Avid basketball player and musician (guitar and cello)